

AB POWER ENGINEERING

Engr. Alexis T. Belonio, MS, ABE, ASEAN Engineer
*Freelance Consultant and Technology Developer; Former Senior Research Fellow (PhilRice)
and Associate Professor (Central Philippine University)*
Member: PSABE, ASABE, ASME

Copyright 2025: Alexis T. Belonio

1

DRAFT ANIMAL POWER

Copyright 2025: Alexis T. Belonio

2

1. Large domesticated four-footed animals used as power for agriculture.

- a. Dairy animals
- ☒ b. Draft animal
- c. Game animals
- d. None of the above

2. Draft animals are basically used for ____.

- a. pulling implements
- b. transport
- c. stationary operation like milling
- ☒ d. All of the above
- e. None of the above

3. Which of the following is not a draft animal?

- a. Oxen
- b. Water buffalo
- c. Horses
- ☒ d. None of the above

4. An object used to optimize the energy potential of an animal in exerting force for transport, pulling of cart or implement, or for driving animal-powered gear.

- a. Implement or cart
- ☒ b. Harness
- c. Person attending the animal
- d. None of the above

5. Draft animals are harnessed together for the reason that ____.

- a. they are easier to manage during tilling operation
- b. it is more comfortable for the animal to work when they are harnessed together
- ☒ c. it matches the loading capacity or the load of the implement
- d. None of the above

6. The smallest draft animal.

- a. Cow
- b. Horse
- ☒ c. Donkey
- d. None of the above

Ref. 1-102 bovines

10-12 horses

animal force = 672 lbs body weight

Copyright 2025: Alexis T. Belonio

3

7. A device that converts energy of animal to useful work.

- a. Head yoke
- b. Breast strap
- ☒ c. Harness
- d. None of the above

8. Harness for animals are designed and built properly to ____.

- a. reduce its cost
- ☒ b. improve the performance efficiency of the animals in pulling the load
- c. make it better looking during tilling
- d. All of the above

9. Good harnessing of draft animals results in a/an ____.

- a. healthy animals
- ☒ b. improved animal-implement performance
- c. improved feeding performance of animals
- d. None of the above

10. Animals that are harnessed together can provide ____ output than when they are used to work individually.

- ☒ a. lesser
- b. more
- c. the same
- d. None of the above

11. The commonly used harness for bovines.

- a. Collar
- ☒ b. Yoke
- c. Head yoke
- d. None of the above

12. Animals is ____ powerful than human.

- ☒ a. 1 to 4 times
- b. 5 to 10 times
- c. 11 to 15 times
- d. None of the above

Copyright 2025: Alexis T. Belonio

4

13. Draft animals that are harnessed in multiple numbers.

- a. Buffalo
- b. Cow
- c. Horses
- d. All of the above

14. The approximate pull of draft animals.

- a. 10 to 30 kg
- b. 30 to 80 kg
- c. 80 to 120 kg
- d. 120 to 200 kg
- e. All of the above

15. What is the approximate pull of a carabao that weighs 1000 kilogram?

- a. 100 kg
- b. 150 kg
- c. 200 kg
- d. None of the above

16. If two animals having 0.75 hp each are harnessed together, the pulling output of the two animals is ____.

- a. greater than 1.5 hp
- b. equal to 1.5 hp
- c. less than 1.5 hp
- d. None of the above

17. The advantage of draft animals over power tiller.

- a. They can feed themselves.
- b. They can maintain themselves.
- c. They can reproduce themselves.
- d. All of the above
- e. None of the above

18. Which of the following is not applied for animal power?

- a. Plowing and harrowing
- b. Transport
- c. Water lifting
- d. Threshing
- e. None of the above

Copyright 2025: Alexis T. Belonio

5

19. A 800-kg water buffalo is pulling a plow at a speed of 0.9 m/s, what is the draft power of the animal?

- a. 0.55 hp
- b. 0.96 hp
- c. 1.12 hp
- d. None of the above

$$800 \text{ kg} \times 0.9 = 720 \text{ kg/s}$$

$$800 \text{ kg} \times 1.0 = 800$$

20. If the animal above is pulling a 10-cm wide plow, how long will the animal travel in order to finish one hectare?

- a. 75 km
- b. 100 km
- c. 125 km
- d. None of the above

$$\frac{10000 \text{ m}^2}{10 \text{ cm}} = 1000$$

21. The commonly used equipment for transport by putting it on the back of an animal for carrying the product.

- a. Cart
- b. Wooden sledge
- c. Saddles
- d. All of the above

22. Which of the following is not a transport animal?

- a. Horse
- b. Donkey
- c. Camel
- d. All of the above
- e. None of the above

23. The characteristic of an animal that affects its performance.

- a. Breed
- b. Weight
- c. Sex
- d. Training
- e. Quality of feed
- f. All of the above

Copyright 2025: Alexis T. Belonio

6

24. The power developed by draft animals.

- a. Less than 1 hp
- b. 1 to 1.5 hp
- c. 1.5 to 2 hp
- d. All of the above

25. Compute the power developed by a 600-kg horse pulling an 80-kg load at 0.9 m/sec speed.

- a. 0.94 hp
- b. 0.87 hp
- c. 0.99 hp
- d. None of the above

$$80 \text{ kg} \times 0.9 = 72 \text{ kg m/s}$$

26. Which of the following animal is stronger?

- a. Cow
- b. Carabao
- c. Horse
- d. All of the above

27. The performance efficiency of bovine animals.

- a. 5-8%
- b. 9-10 %
- c. 11 – 15%
- d. d. None of the above

28. The performance efficiency of horses.

- a. 9-10%
- b. 10-12%
- c. 12 -15%
- d. All of the above

$$\text{human} = 20\%$$

29. The working period of horses.

- a. 5 – 8 hours
- b. 8 – 10 hours
- c. 12 – 16 hours
- d. None of the above

Copyright 2025: Alexis T. Belonio

7

30. Which of the following is not an animal transport equipment?

- a. Saddles
- b. Wooden Sledges
- c. Cart
- d. None of the above

31. The average speed of draft animals.

- a. Less than 1 m/s
- b. 1 to 2 m/s
- c. More than 2 m/s
- d. None of the above

32. Approximate number of days required for one animal working for 5 to 6 hours per day to finish plowing 1 hectare?

- a. 2 to 6 days
- b. 6 to 9 days
- c. 10 to 12 days
- d. None of the above

33. Six (6) horses can provide a pulling load of 60 kg per animal. How much draft can the 6 horses provide in pulling a load?

- a. 167 kg
- b. 229 kg
- c. 304 kg
- d. None of the above

$$6 \times 60 \times 3 \frac{8}{6} = 229$$

$$38 \text{ for 6 horses}$$

Copyright 2025: Alexis T. Belonio

8

GOD BLESS!!!

Copyright 2025: Alexis T. Belonio